Assignment II

SDG-Based Questions – CO2

Course Code	ITA06
Course Name	Machine Learning
Faculty Handling	Dr.K.Vivek Balaji
Course Outcome	CO2 – Neural Networks and Genetic Algorithms

Instructions: Answer any question(s) as directed by the faculty. Each question maps a Neural Network / Genetic Algorithm concept to a real-world application aligned with the United Nations Sustainable Development Goals (SDGs).

Q.No	Question
1	Design a single-layer perceptron that classifies water samples as 'safe' or 'unsafe' for drinking based on pH, turbidity, and dissolved oxygen levels. Explain the weight update rule you would use. (SDG 6 – Clean Water and Sanitation)
2	A perceptron is trained to classify farmland soil as 'fertile' or 'degraded' using nitrogen, phosphorus, and potassium content as inputs. Derive the perceptron learning rule and explain convergence conditions. (SDG 2 – Zero Hunger)
3	Explain why a single perceptron cannot solve the problem of classifying households as 'energy poor' or 'energy secure' when the decision boundary is non-linear. Illustrate with a suitable example. (SDG 7 – Affordable and Clean Energy)
4	Using the perceptron algorithm, design a binary classifier that predicts whether a patient is at risk of malnutrition based on BMI and dietary intake. Show two iterations of weight updates with sample data. (SDG 2 – Zero Hunger)
5	Discuss the representational limitations of perceptrons in the context of classifying air quality as 'hazardous' or 'safe' when pollutant interactions are non-linear. (SDG 11 – Sustainable Cities and Communities)
6	A perceptron-based system is used to flag illegal deforestation activity from satellite pixel data (vegetation index vs. bare soil index). Explain the perceptron representation and its threshold function. (SDG 15 – Life on Land)
7	Explain the geometric interpretation of a perceptron's decision boundary using an example of classifying coastal water samples as 'polluted' or 'clean'. (SDG 14 – Life Below Water)
8	Compare the perceptron convergence theorem with real-world limitations when applied to predicting flood-risk zones from rainfall and elevation data. (SDG 13 – Climate Action)
9	Represent a neural network architecture (input, hidden, output layers) suitable for predicting crop yield from soil moisture, temperature, and rainfall data. Justify the number of neurons chosen. (SDG 2 – Zero Hunger)
10	Identify and explain three real-world problems in renewable energy forecasting (solar/wind) that make them suitable candidates for neural network representation rather than rule-based systems. (SDG 7 – Affordable and Clean Energy)

Q.No	Question
11	Explain how neural network representation handles noisy and incomplete sensor data in a smart water-leakage detection system for urban infrastructure. (SDG 6 – Clean Water and Sanitation)
12	Discuss the appropriate problem characteristics (as per Mitchell's criteria) that justify using neural networks for early detection of disease outbreaks from epidemiological data. (SDG 3 – Good Health and Well-being)
13	Design the input and output representation of a neural network intended to predict student dropout risk in under-resourced schools. Justify your feature encoding choices. (SDG 4 – Quality Education)
14	Explain the challenges in representing multi-modal data (satellite imagery + weather data) as neural network inputs for wildfire prediction systems. (SDG 15 – Life on Land)
15	Design a multilayer perceptron architecture to classify recyclable vs non-recyclable waste from sensor-based material composition data. Specify the number of hidden layers and activation functions used. (SDG 12 – Responsible Consumption and Production)
16	Explain how a multilayer perceptron overcomes the linear separability limitation of a single perceptron, using the example of predicting energy consumption patterns in smart grids. (SDG 7 – Affordable and Clean Energy)
17	A multilayer perceptron is used to predict maternal health risk levels (low/medium/high) from blood pressure, blood sugar, and age. Draw and explain the network architecture with appropriate activation functions. (SDG 3 – Good Health and Well-being)
18	Explain the role of hidden layers in a multilayer perceptron used to forecast urban air pollution levels from traffic density and weather variables. (SDG 11 – Sustainable Cities and Communities)
19	Design an MLP-based model to classify groundwater contamination levels using multiple chemical indicators as input features. Justify the choice of output layer activation function. (SDG 6 – Clean Water and Sanitation)
20	Derive the backpropagation weight update rule for a neural network predicting drought severity from rainfall, soil moisture, and temperature data. (SDG 13 – Climate Action)
21	Explain the vanishing gradient problem in backpropagation and its impact on a deep network trained to detect early signs of crop disease from leaf images. (SDG 2 – Zero Hunger)
22	Trace the forward pass and backward pass of backpropagation for a simple 2-input, 1-hidden-layer network predicting whether a slum household has access to sanitation facilities. (SDG 6 – Clean Water and Sanitation)
23	Discuss how the learning rate parameter in backpropagation affects convergence speed and accuracy when training a model to predict renewable energy output from weather forecasts. (SDG 7 – Affordable and Clean Energy)
24	Explain the role of the error (delta) term in backpropagation using an example of a network trained to classify malnutrition risk in children under five. (SDG 2 – Zero Hunger)

Q.No	Question
25	Compare batch, stochastic, and mini-batch gradient descent in the context of training a backpropagation network on large-scale climate sensor data for extreme weather prediction. (SDG 13 – Climate Action)
26	Explain how deep neural networks improve upon shallow networks for detecting illegal fishing vessels from satellite imagery, referencing feature hierarchy learning. (SDG 14 – Life Below Water)
27	Discuss the role of convolutional and pooling layers (conceptually) in a deep network designed to classify satellite images of deforested vs forested land. (SDG 15 – Life on Land)
28	Explain overfitting in deep neural networks and describe two regularization techniques suitable for a small dataset predicting rare disease outbreaks in low-resource regions. (SDG 3 – Good Health and Well-being)
29	Discuss how transfer learning in deep neural networks can accelerate the development of models for identifying safe drinking water sources in data-scarce rural regions. (SDG 6 – Clean Water and Sanitation)
30	Explain why deep neural networks are preferred over shallow MLPs for processing high-resolution thermal imagery in urban heat island analysis. (SDG 11 – Sustainable Cities and Communities)
31	Formulate a genetic algorithm (chromosome encoding, fitness function, selection method) to optimize the placement of solar panels in a smart city for maximum energy yield. (SDG 7 – Affordable and Clean Energy)
32	Design a genetic algorithm to optimize waste-collection vehicle routing in a city to minimize fuel consumption and emissions. Define the fitness function clearly. (SDG 11 – Sustainable Cities and Communities)
33	Explain the crossover and mutation operators of a genetic algorithm used to optimize irrigation schedules for water conservation in agriculture. (SDG 6 – Clean Water and Sanitation)
34	A genetic algorithm is used to optimize crop rotation plans to maximize yield while preserving soil fertility. Describe the population representation and selection strategy. (SDG 2 – Zero Hunger)
35	Discuss how genetic algorithms can be used to optimize the design of low-cost water filtration systems for developing regions, including a suitable fitness function. (SDG 6 – Clean Water and Sanitation)
36	Explain the concept of hypothesis space search in the context of a genetic algorithm evolving neural network weights for predicting famine risk in vulnerable regions. (SDG 2 – Zero Hunger)
37	Compare hypothesis space search in genetic algorithms versus gradient-based search in backpropagation, using flood prediction modeling as a case study. (SDG 13 – Climate Action)

Q.No	Question
38	Explain how genetic programming could be used to automatically evolve a predictive formula for estimating renewable energy demand in off-grid rural communities. (SDG 7 – Affordable and Clean Energy)
39	Discuss the representation of candidate programs (as parse trees) in genetic programming applied to evolving decision rules for sustainable forest resource management. (SDG 15 – Life on Land)
40	Compare Lamarckian and Baldwin effect models of evolution and learning, and discuss their relevance to evolving adaptive neural controllers for low-cost prosthetic devices in developing countries. (SDG 3 – Good Health and Well-being)

3) 1. Problem Understanding

A **single-layer perceptron** is a simple neural network that performs **binary classification** by separating data using a **linear decision boundary** (a straight line in 2D, a plane in higher dimensions). It can classify data correctly only when the two classes are **linearly separable**.

In the context of **SDG 7 (Affordable and Clean Energy)**, households may be classified as:

- **Energy Secure (1):** Reliable electricity, affordable energy cost, and sufficient daily energy supply.
- **Energy Poor (0):** Limited electricity access, high energy cost, and insufficient energy availability.

However, real-world energy data often exhibit **non-linear relationships**, making it impossible for a single perceptron to classify all households accurately.

2. Theory

A perceptron computes the weighted sum of inputs:

y={1,if w1x1+w2x2+⋯+wnxn+b≥00,otherwisey= \begin{cases} 1,& \text{if } w_1x_1+w_2x_2+\cdots+w_nx_n+b\ge0\\ 0,& \text{otherwise} \end{cases}y={1,0,if w1x1+w2x2+⋯+wnxn+b≥0otherwise

where:

- xix_ixi = input features
- wiw_iwi = weights
- bbb = bias
- yyy = output class

The decision boundary is

wTx+b=0w^Tx+b=0wTx+b=0

which is always **linear**.

Therefore, if the data require a curved or complex boundary, a single perceptron cannot classify them correctly.

3. Example (Energy Poverty Classification)

Suppose classification depends on two features:

- **x_1:** Monthly Electricity Consumption
- **x_2:** Monthly Energy Cost

Household	Electricity Consumption	Energy Cost	Class
A	Low	Low	Energy Poor
B	High	High	Energy Poor
C	Low	High	Energy Secure
D	High	Low	Energy Secure

This produces an **XOR-like pattern**.

```
Energy Cost

High         C (Secure)       B (Poor)

Low          A (Poor)         D (Secure)

             Low              High
          Electricity Consumption
```

No single straight line can separate the "Energy Secure" and "Energy Poor" households.

Hence, the perceptron fails.

4. Why a Single Perceptron Fails

A single perceptron assumes one linear boundary.

For XOR-type data:

- One class appears in opposite corners.
- The other class occupies the remaining corners.

Since no straight line separates these classes, the perceptron never reaches zero classification error, regardless of weight updates.

Therefore:

- Training does not converge.

- Classification accuracy remains low.
- Complex relationships between features cannot be learned.

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5. Solution

A **Multi-Layer Perceptron (MLP)** overcomes this limitation.

It includes:

- Input Layer
- One or more Hidden Layers
- Output Layer

Hidden neurons learn intermediate features, allowing the network to create **non-linear decision boundaries**.

```
Inputs
(Electricity Consumption, Energy Cost)
        │
        ▼
   Hidden Layer
      o o o
        │
        ▼
 Output Layer
 Energy Poor / Energy Secure
```

Using activation functions such as **ReLU** or **Sigmoid**, the MLP can correctly classify households with complex energy-use patterns.

6. Advantages of Using MLP

- Learns non-linear relationships.
- Handles complex real-world energy consumption patterns.
- Provides higher prediction accuracy.
- Suitable for smart-grid and renewable energy applications.
- Supports better decision-making for identifying energy-poor households.

7. Conclusion

A single perceptron is effective only for **linearly separable** datasets. Household energy classification often depends on multiple interacting factors, creating **non-linear decision boundaries**. Consequently, a single perceptron cannot solve this problem. A **Multi-Layer Perceptron (MLP)** with hidden layers and non-linear activation functions is required to accurately classify households as **energy poor** or **energy secure**, supporting **SDG 7 – Affordable and Clean Energy**.